

DATA & AI IN ECONOMICS — MISSION PROPOSAL

The Gender Pay Gap at Work

Does Being a Woman Causally Reduce Your Salary?

Group A

Emre Isildar • Mohammadali Vahabi

TU Dortmund — Data and AI in Economics

Motivation & Economic Relevance

Research question

Does gender causally affect base salary among full-time employees — after controlling for job title, seniority, education, age and performance — and how large is the unexplained gap?

- Raw gender pay gaps are well documented, but it is unclear how much reflects discrimination versus occupation, experience, or education differences.
- Disentangling correlation from causation matters directly for equal-pay legislation, HR policy, and corporate pay-equity audits.
- We use causal inference (not just prediction) to isolate the direct effect of gender on pay, net of observable confounders.

Raw (unadjusted) pay gap in our data

\$8,515

Male average \$98,458 vs. Female average \$89,943

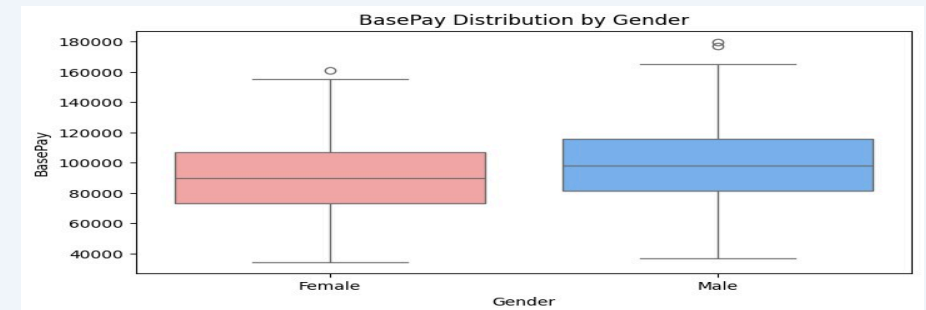


Fig. 1 — BasePay distribution by gender (n = 1,000)

Data & Methodology

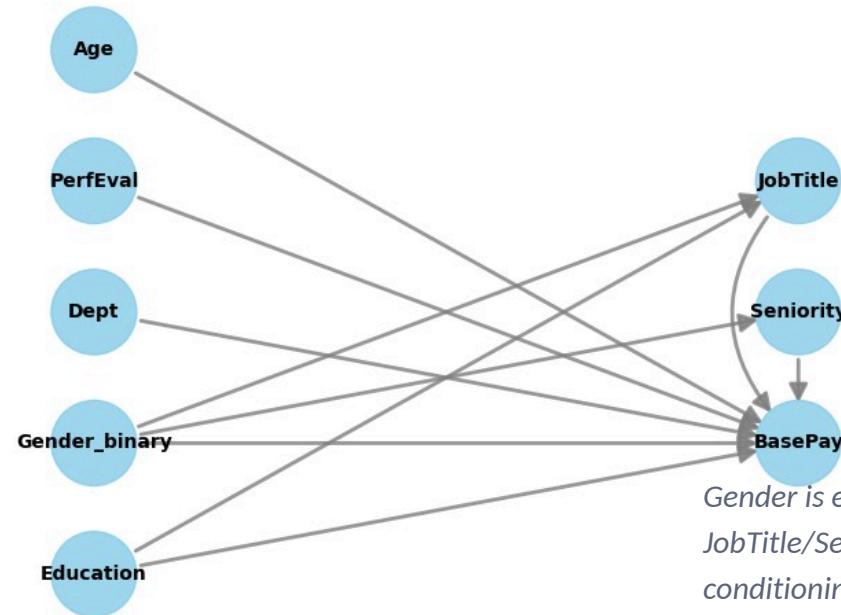
Data

- Glassdoor Gender Pay Gap dataset (Kaggle, CC0)
- n = 1,000 employees, 9 variables
- Gender, Age, Education, Dept, JobTitle, Seniority, PerfEval, BasePay, Bonus
- No missing values; not Germany-specific

Three method blocks

- Causal: DAG + propensity score stratification
- Supervised: Linear / Ridge / Lasso / Tree / Random Forest (tuned)
- Unsupervised: K-Means (K=3) on scaled features

Causal graph (DAG): Gender → Seniority/JobTitle are mediators, not confounders



Gender is excluded from the propensity model alongside JobTitle/Seniority being excluded as mediators — conditioning on a mediator would block the causal pathway (post-treatment bias).

Results — Causal Inference

Adjusted Average Treatment Effect

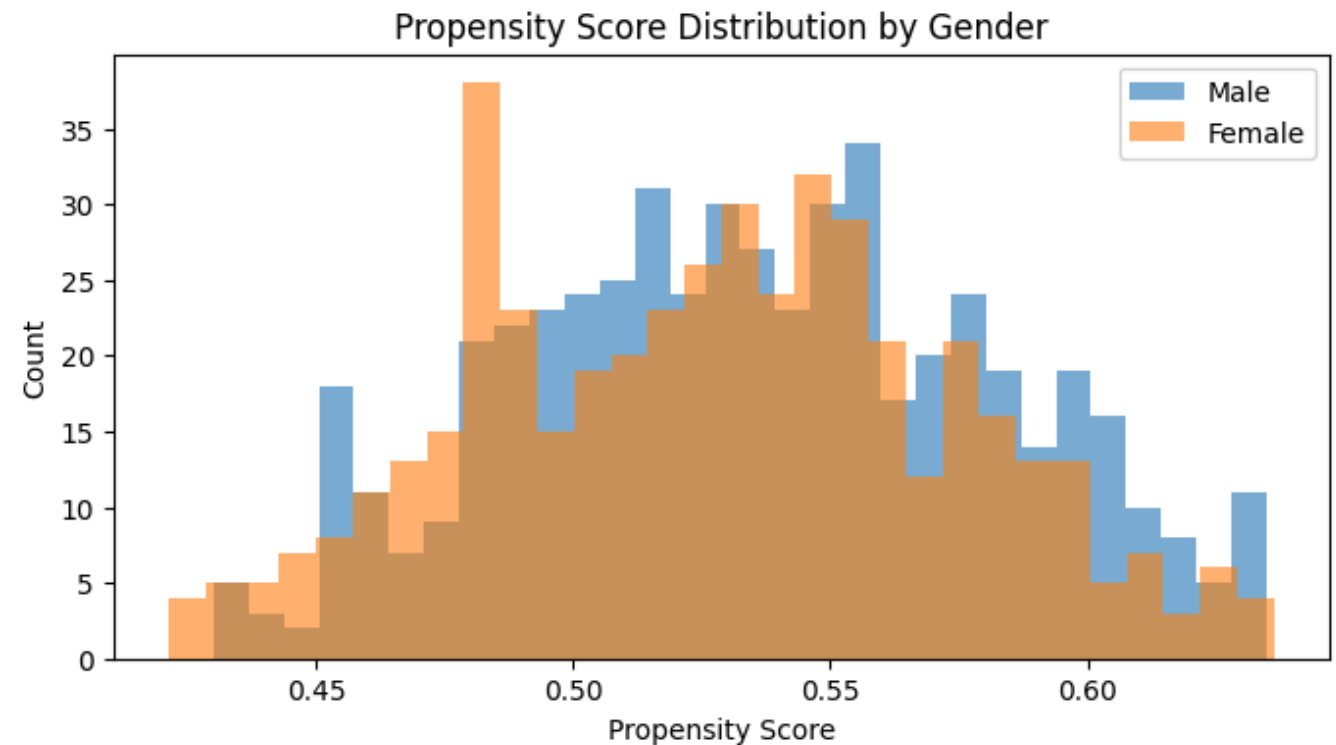
\$9,186

vs. raw (unadjusted) gap of \$8,515

Refutation tests

- Placebo treatment: ATE $\rightarrow$ $-\$1,193$ (≈ 0) ✓
- Random common cause: ATE $\rightarrow$ $\$8,862$ ($\Delta = \$325$) ✓
- Both tests support a reasonably robust causal estimate

Common support: propensity score distribution by gender



Overlapping distributions confirm common support — stratification (5 quintiles) compares like-with-like without discarding observations.

Results — Prediction & Heterogeneous Effects

Supervised learning: Ridge $\approx$ tuned Random Forest

- Ridge: $R^2 \approx 0.863$, RMSE $\approx$ \$9,286 (best RMSE)
- Random Forest (tuned via GridSearchCV): $R^2 \approx 0.887$ (test), CV $R^2 \approx 0.782$
- Near-identical Ridge/RF performance $\Rightarrow$ BasePay is largely linear in observables

Heterogeneous treatment effects (K-Means $\times$ causal)

Cluster	Segment	Pay Gap
0 (n=273)	Senior, high-pay, low perf.	\$8,757
1 (n=316)	Senior, high-pay, high perf.	\$9,223
2 (n=411)	Younger, lower-pay, lower seniority	\$6,390

Gap is smallest among the youngest/lowest-seniority segment and largest among senior segments — consistent with Gender $\rightarrow$ Seniority $\rightarrow$ BasePay.

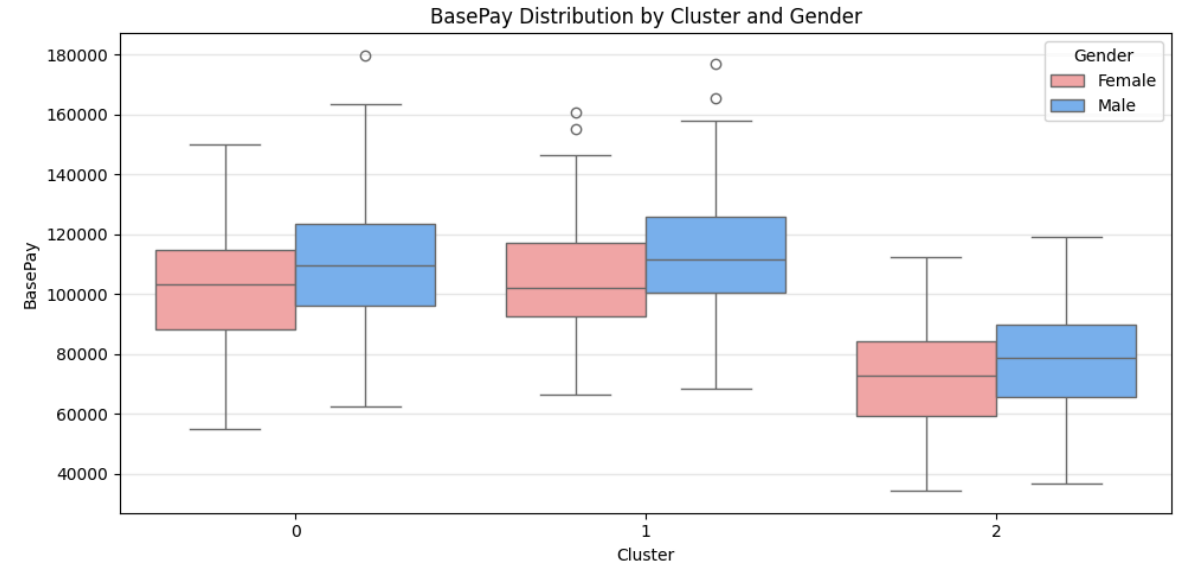


Fig. 2 — BasePay by cluster and gender (StandardScaler applied pre-K-Means)

Limitations & Synthesis

Key limitations

- Synthetic / heavily engineered data (~1,000 rows) — likely reverse-engineers the generator rather than real labor-market dynamics; limits external validity
- Omitted Variable Bias: hours worked, career interruptions, negotiation outcomes, firm characteristics are all unobserved
- Mediator-only specification: \$9,186 ATE is the direct effect only — excludes the indirect effect via promotions/seniority access
- Sample size (~1,000): Random Forest underperforms a simple Ridge even after tuning
- Modest silhouette scores (0.27–0.33): clusters are an exploratory approximation, not ground truth

What each lens reveals that the others miss

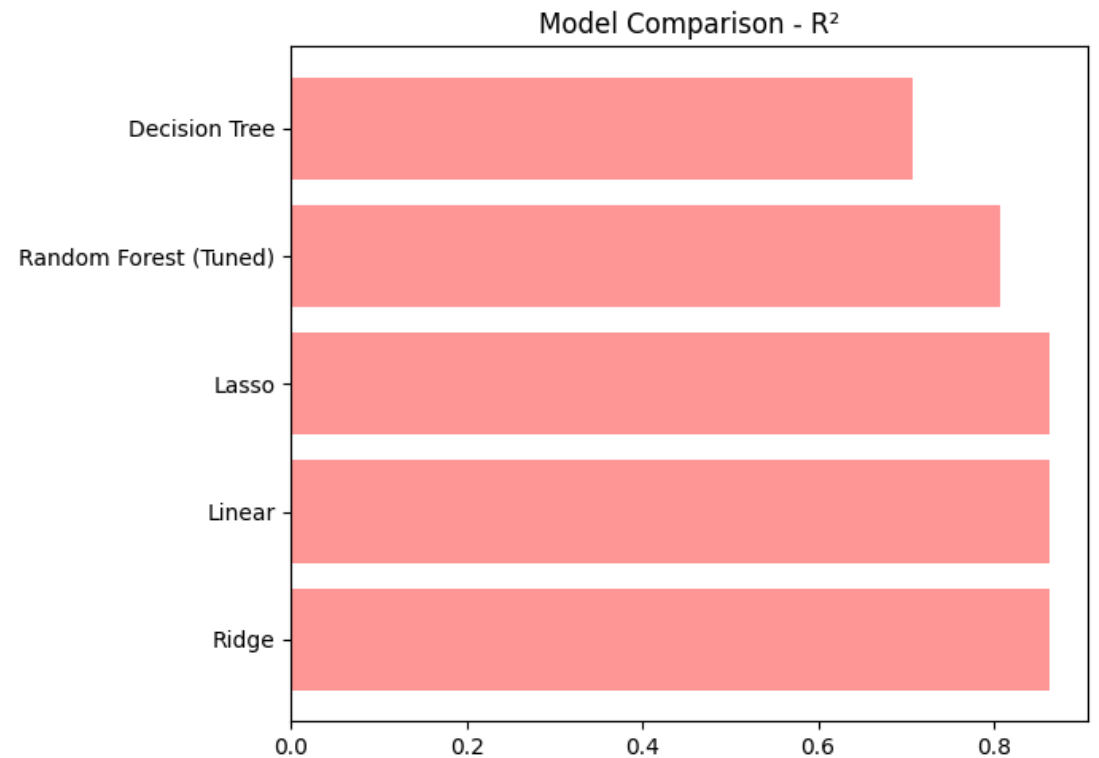
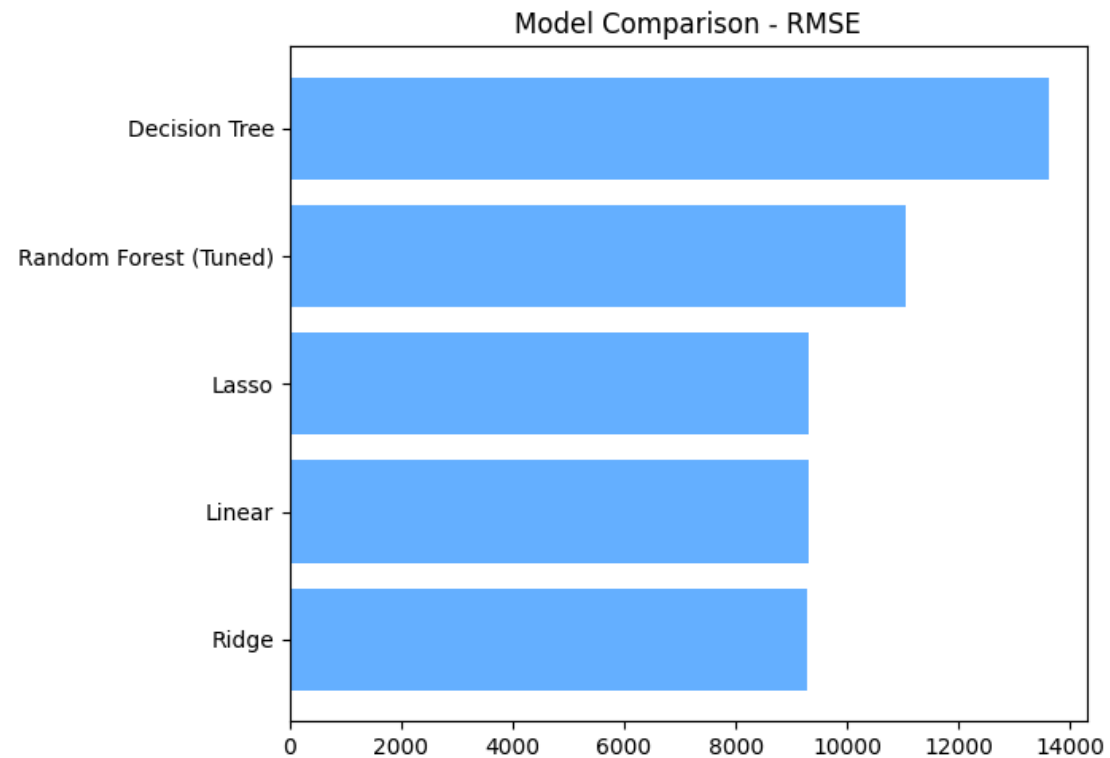
Causal: isolates the direct, average effect — but gives no individual prediction or subgroup variation.

Supervised: predicts pay accurately and ranks features — but importance is associative, not causal (cannot tell mediator from confounder).

Unsupervised: reveals latent employee segments invisible to the other two — and, combined with the causal model, exposes effect heterogeneity a single pooled ATE hides.

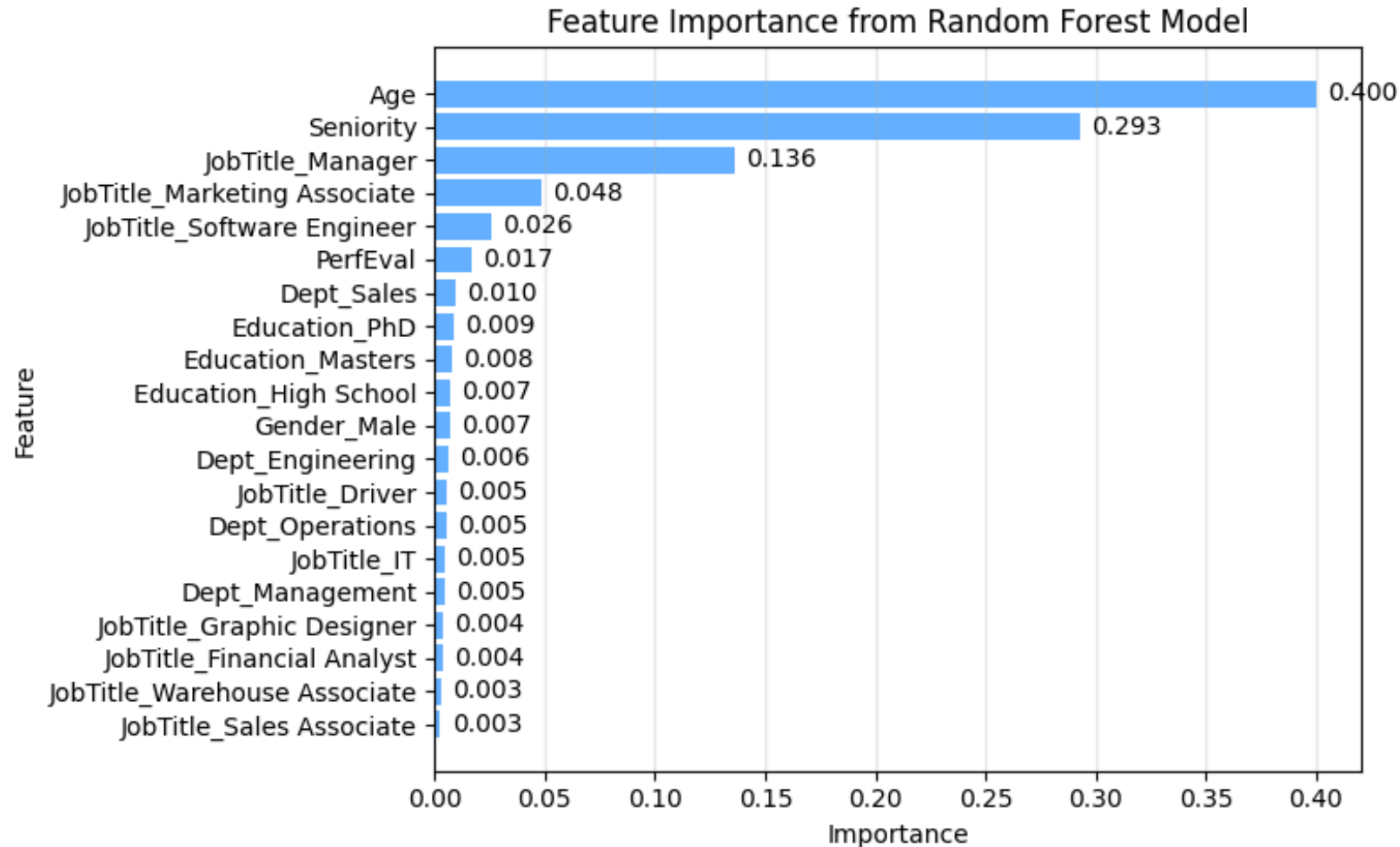
Conclusion: a consistent ~\$9,186 gender pay gap emerges across all three lenses, larger for senior employees — but it should be read as a lower-bound, internally-consistent estimate within this dataset, not a generalizable real-world claim.

Backup — Supervised Model Comparison

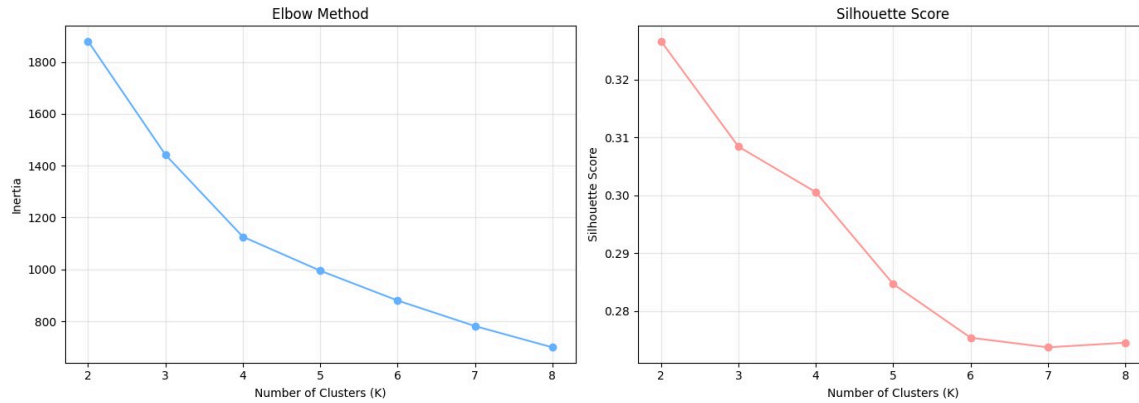


Linear / Ridge / Lasso $\approx$ tuned Random Forest in R^2 and RMSE — Decision Tree clearly weakest. Ridge selected as the best linear baseline (lowest RMSE).

Backup — Random Forest Feature Importance



Backup — K-Means Diagnostics



K=2 maximizes silhouette numerically (0.327) but K=3 (0.308) chosen for interpretable, career-stage-aligned segments.

